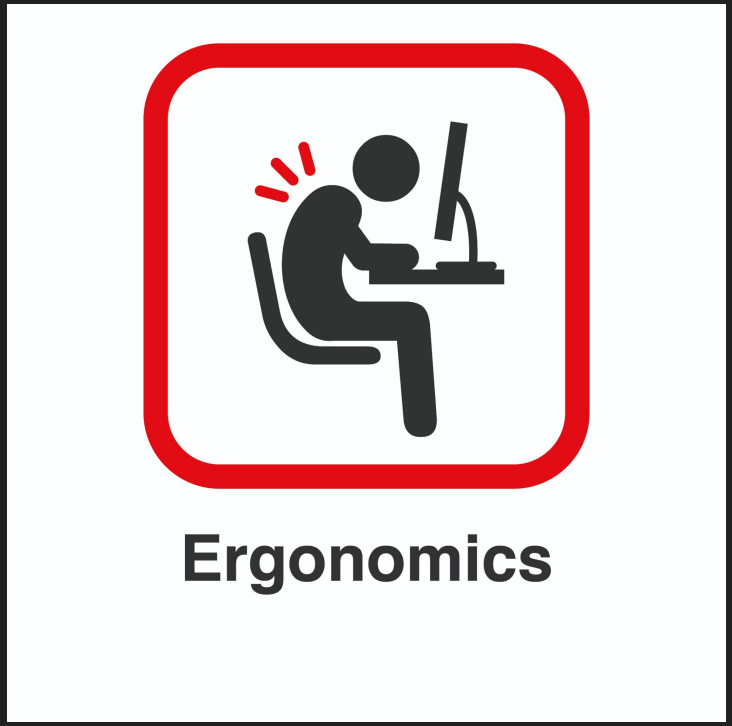
Computer Safety

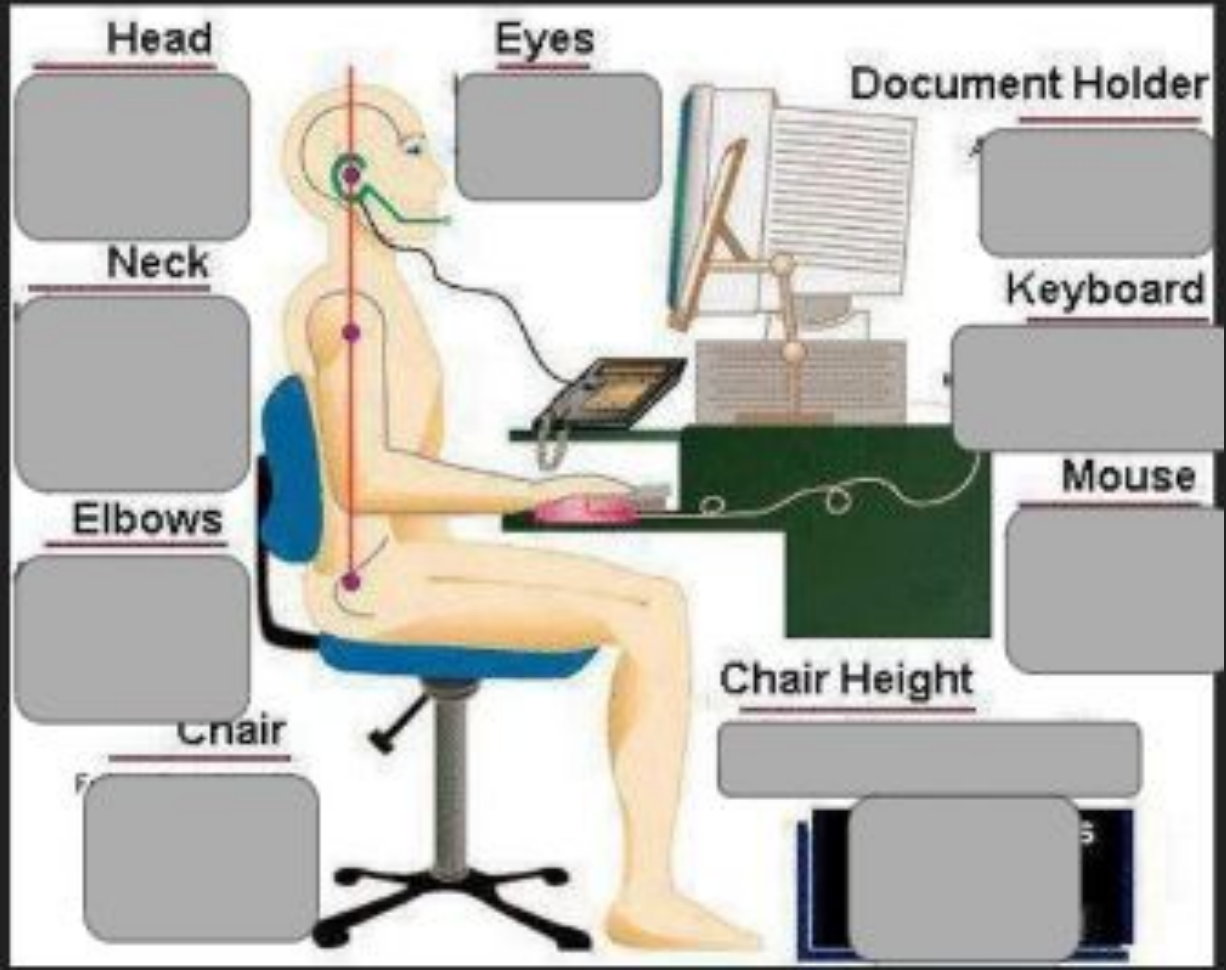
Inside and Outside

Ergonomics

Refers to the study of the workplace and how it affects human health



The Computer Workstation Ergonomics



Computer Construction Safety

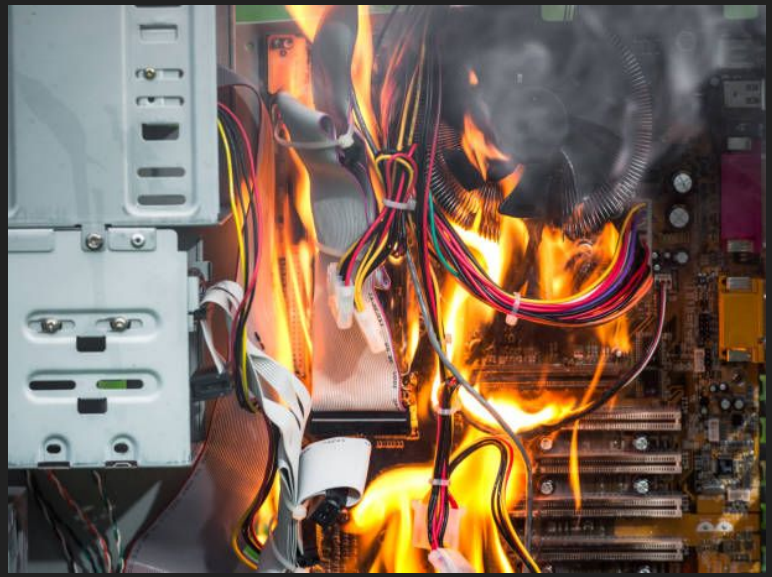
When working with a computer there are safety considerations for you and the computer.



Short CIrcuit

The biggest concern for the computer is a

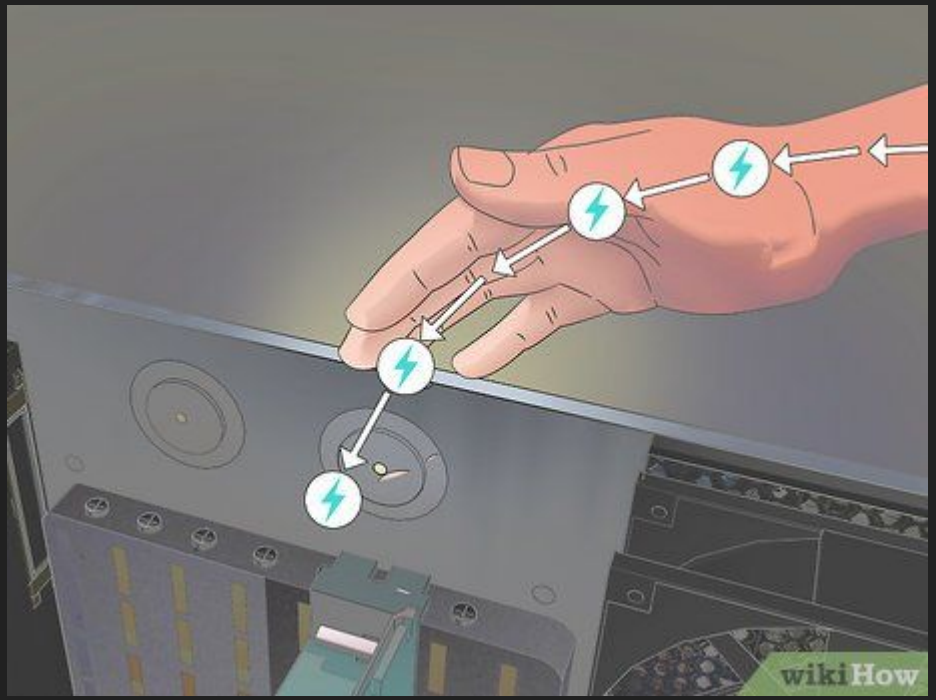
short circuit. This is caused when electricity has no resistance and is free to flow without restriction. It can be caused by supplying a large amount of static electricity coming from your body.



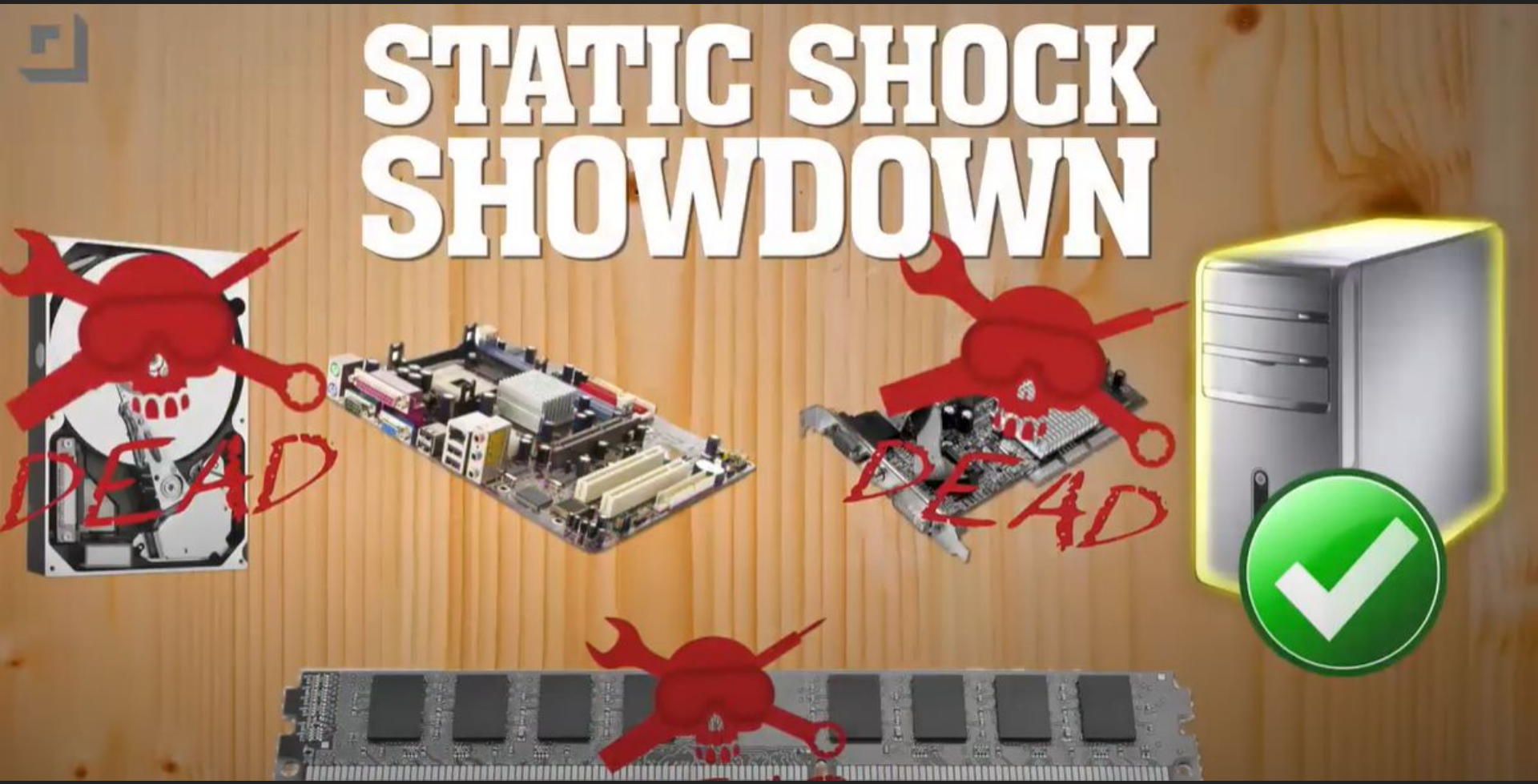
Discharge Static Electricity

1. Touch a metal portion of the computer case
2. Wear an anti-static wrist strap





https://www.youtube.com/watch?v=AhlyMdiLc0E

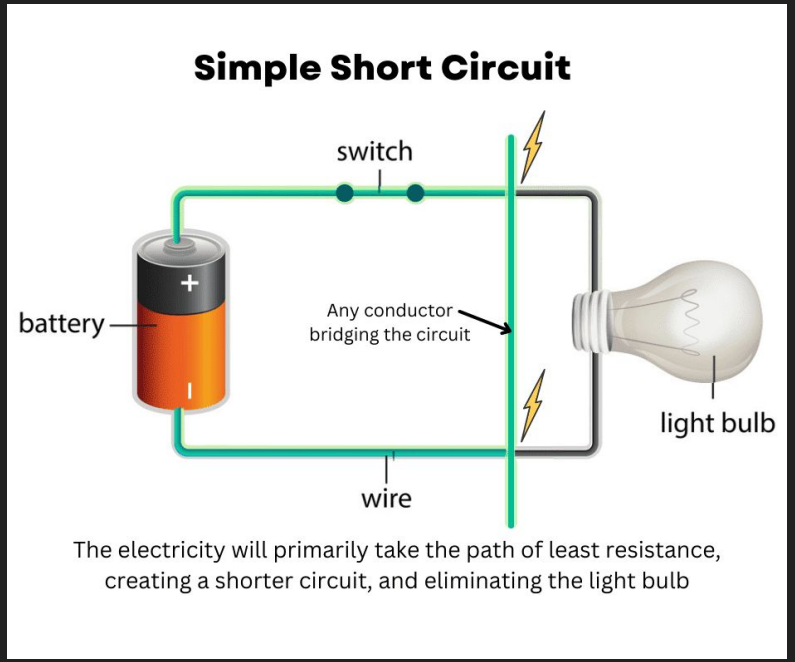


https://www.youtube.com/shorts/zGMyQuNv1_E?feature=share

Short Circuit

In this example a wire or metal object is placed across the circuit giving flowing electrons a free path back to the negative or ground side of a circuit. Normally the electrons would have to go through the bulb which slows it down enough to not cause a fire hazard.

You can not create a short circuit this way if there is no power in your circuit so always make sure to disconnect the computer's power supply before working on the inside of it.



Liquid Hazards

Do not drink or have any liquids near your computer as liquids are great conductors of electricity and can easily create a short circuit.



Disconnecting a Computer’s Power Supply

Always handle a power cable by the plug and not the cable. Why?

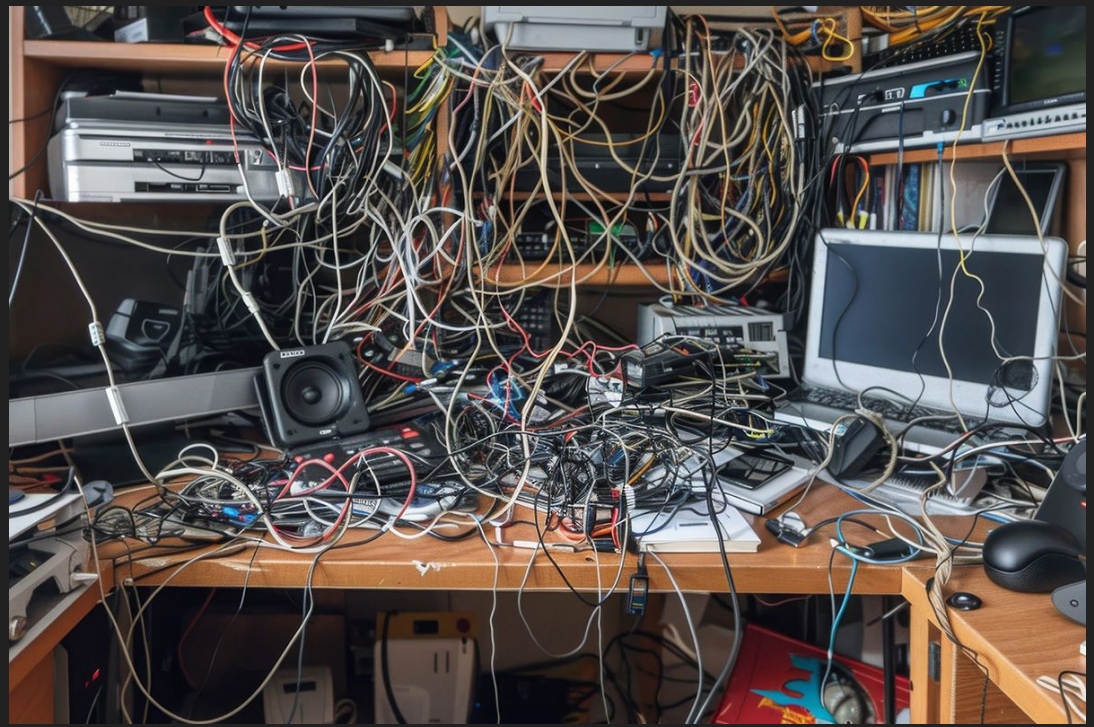
Because pulling the cable may damage the internal wires, causing poor contact, short circuits, or even electric leakage. Handling by the plug ensures the stability of the connection, guarantees electrical safety, and extends the service life of the power cable.



Neat Cabling

Cables that are not neatly arranged can cause a short circuit/fire hazard.

Regularly inspect all cables for damage and wear. Replace worn or damaged cables. Report all cabling problems to your teacher immediatel



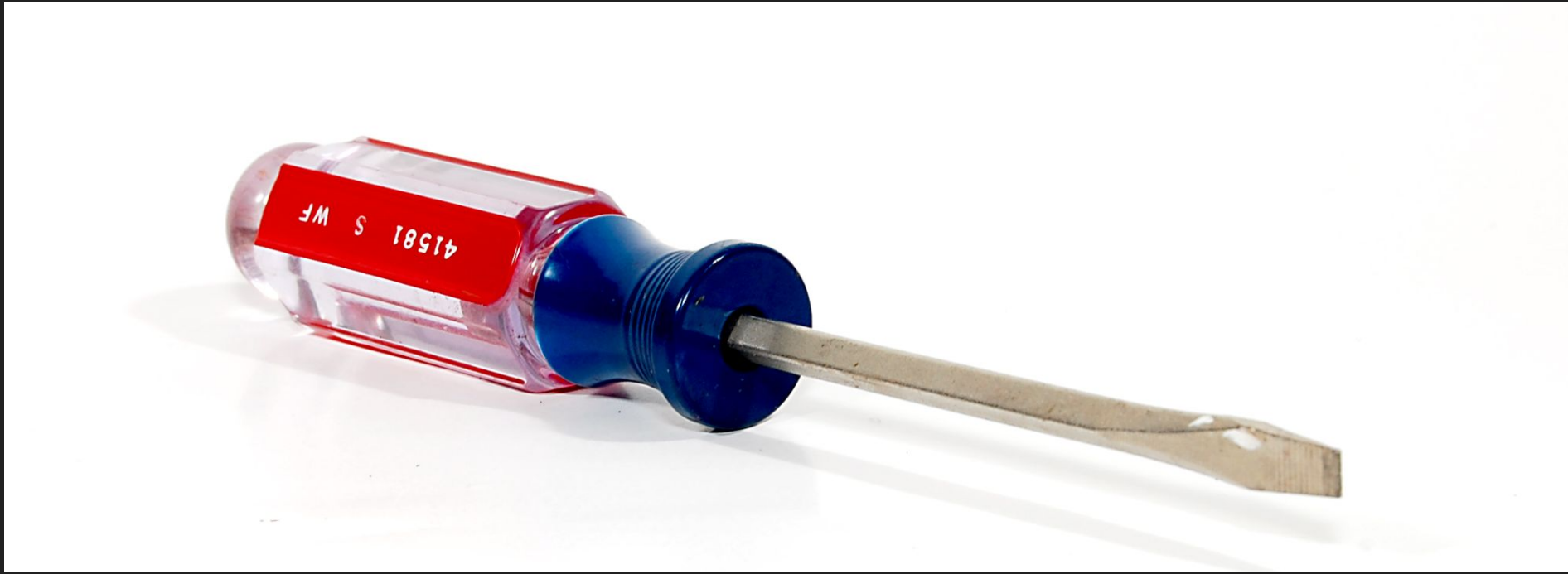
Computer Tools

Phillips Screwdriver



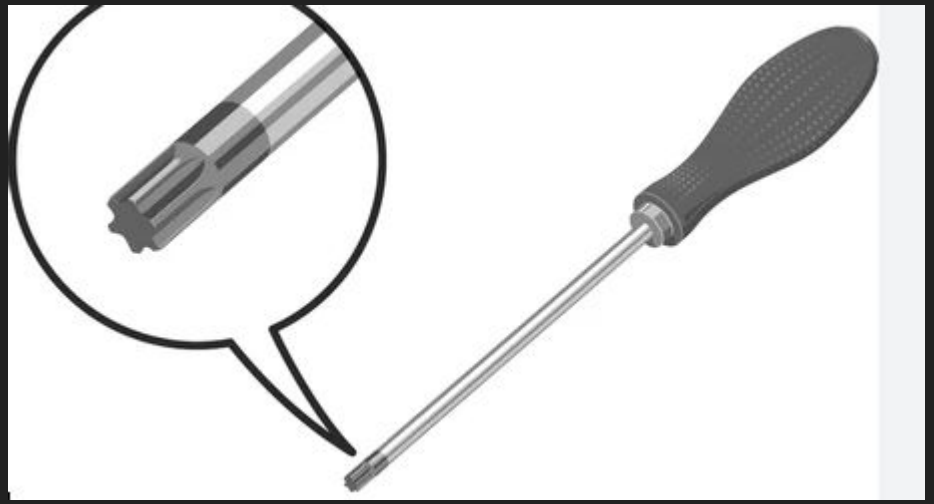
Computer Tools

Flathead Screwdriver



Computer Tools

Torx Screwdriver



Computer Tools

Needle-Nose Pliers



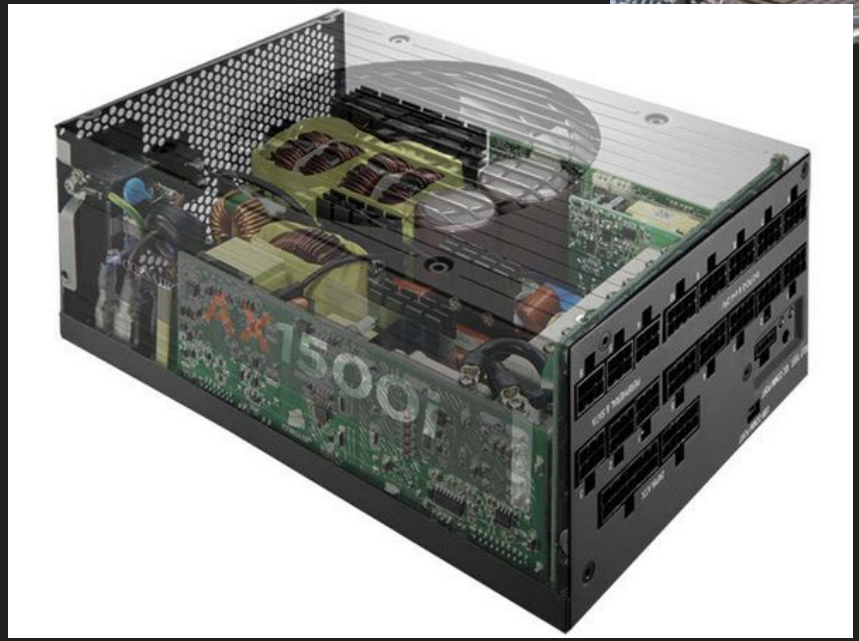
Tool Safety

Never force a tool in your and someone else’s direction as it can slip and hit you or someone else. Never force the pressure applied to a tool beyond a reasonable amount as you can damage a part.

Capacitors

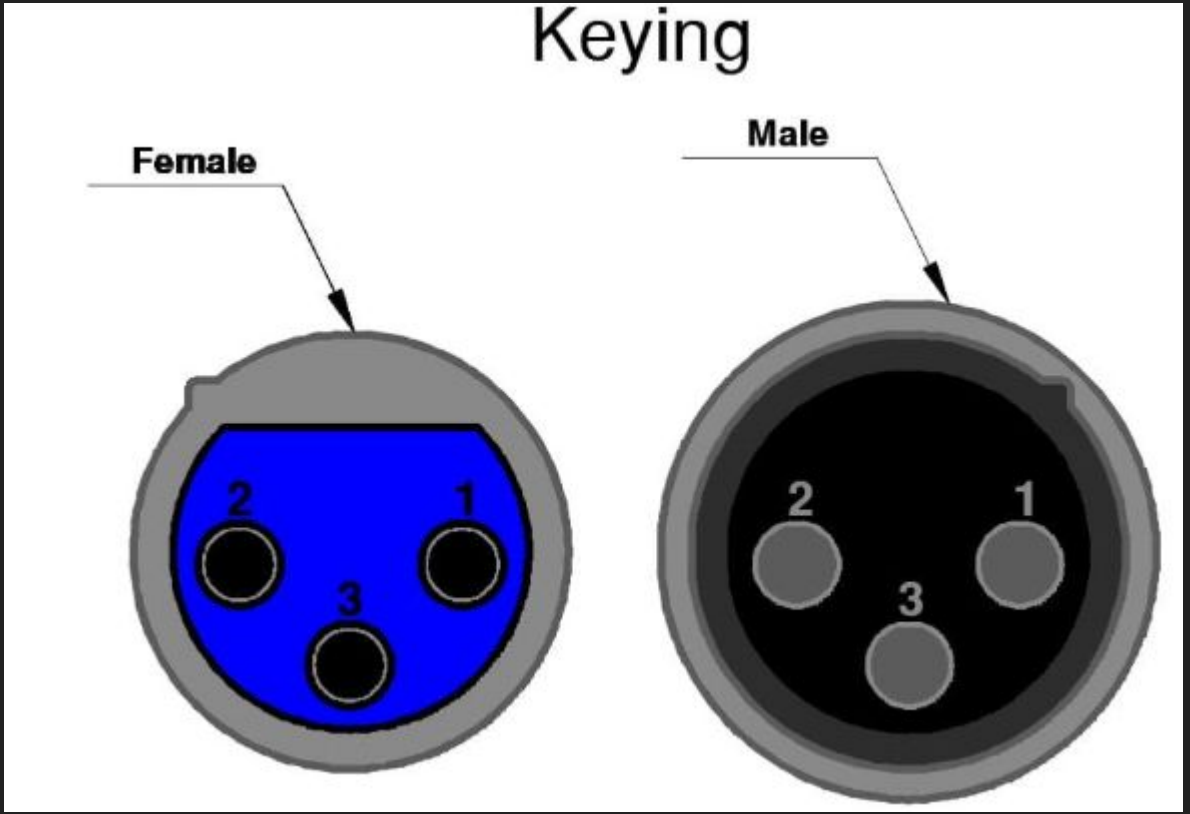
Retain electric even after power supply is cut off. Large ones are found in the PSU. Can be dangerous if touched, particularly with a metal or conductive object. DO NOT STICK ANYTHING INSIDE A PSU AT ANYTIME.





Keying

Parts that connect together are designed such that they can only connect in one way. Often times one part is referred to as the male connector and the other is the female connector. Forcing two connectors together in the wrong way can damage them and possibly the whole computer.



Questions and Exercises

1. Explain the reason for each of the following computer workstation safety considerations:

   Always listen to and follow instructions. Do not operate any equipment unless you have the teacher's permission.

   To ensure safe and correct equipment use, avoiding damage or hazards.

   Prevent accidents and cable damage by allowing sufficient cable length for all planned equipment movement.
   To prevent cable damage and accidents during equipment movement.

   Be aware of unusually warm or hot outlets.To detect electrical issues early and avoid fires or shocks.

   Wear rubber-soled shoes.To insulate against electric shock.

   Relieve long periods of computer use with breaks involving stretching and movement.
   To prevent physical strain from prolonged computer use.

2. At times you will be lifting heavy equipment. Explain how to properly do this to prevent injuries, particularly those to your back.When lifting heavy equipment, squat with bent knees (not bent back), keep the load close to your body, lift with leg muscles, avoid twisting, and ask for help if it's too heavy.

3. Find and list the URLs to locations where you can buy each of the tools discussed in this slideshow.

4. Look at a picture of a SATA data connector on a motherboard (sometimes called header) and the end of a SATA data cable. Explain the keying involved.

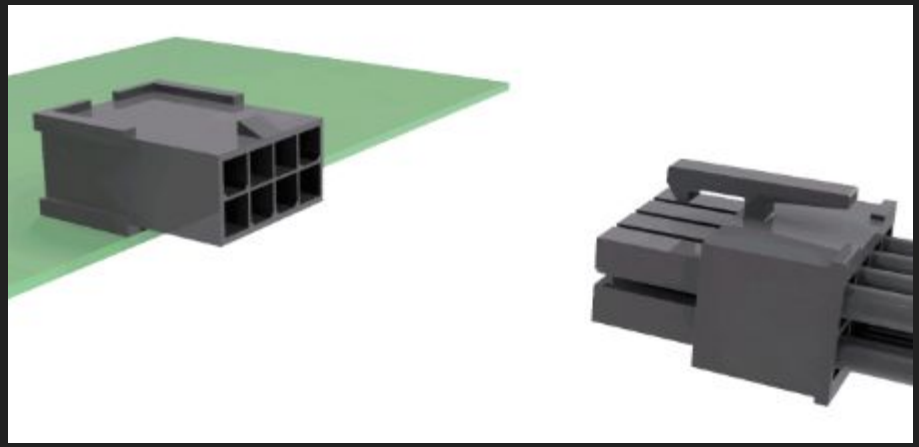
SATA data connectors (on motherboard header and cable end) have a key (protrusion) and matching notch. They only fit one way, preventing reverse insertion.

5. Explain the difference between a SATA data cable key and a SATA power cable key.SATA data cable key: Single key, only prevents reverse insertion.

SATA power cable key: Multiple keys, also distinguish voltage pins to prevent wrong power connection.

6. Which is the female and male connector in the following image:

Female: Left; Male: Right.



7. What is the indication of how these two connectors go together?

The male connector inserts into the female one, matching in shape for connection.

8. Find and list 3 youtube videos that discuss safety as it pertains to working with computers.https://youtube.com/shorts/gfAY2hGnihM?si=03VHiILYfN2pqQEL

https://youtube.com/shorts/1a1_hUhSaqw?si=n0dwuNC2OqUMwm6U
https://youtube.com/watch?v=6mMZFoXbKqI&si=UID49bg2NPBeUe7S

9. Find an embed a picture of an ergonomic keyboard and explain its ergonomic features.

Layout: Split/arcuate design (mimicking natural hand posture), special key arrangements (e.g., Alice layout).

Wrist Support: Built-in wrist rests for comfort and reduced strain.



Key Design: Ergonomic keycaps, optimized key travel and force for natural finger movement.

10. Do the same for a mouse.Grip Adaptation: Supports vertical/prone grips, fits palm curve, reduces carpal tunnel risk.

Control Optimization: Multi-side buttons (customizable), soft-trigger keys/scroll wheel, less finger fatigue.

